1 The held rough error and the closure beyond the lazy regime

Proposition 20 says where every expansion about θ_0 stops: a rough direction is fitted only after a displacement of order the analyticity radius R, so no series in $\theta - \theta_0$ describes the stretch of training in which the rough part of the error is removed. This section changes the expansion variable. The one-time theory of Part I bounds the eigenvalue of the kernel on a rough direction by the peak feature roughness, $\lambda_k R(v_k)^2 \le \sigma_R^2$ (Corollary **??**), and Proposition 15 gives the exact rate of σ_R along training. Together they say that the part of the error that a roughness function counts as rough can lose energy only through one scalar function of time, $\sigma_R(t)$; and the a priori bound on the growth of σ_R then gives a floor on the rough part of the error that holds at every time, with no ball and no horizon.

One point governs everything below and is the reason the section is written as it is. A roughness function is a dual variable for a lower bound: every seminorm gives one, and the supremum over all of them is attained by a single rank-one functional built from the current error and the current kernel, the slow part $E_\gamma = \gamma(K + \gamma I)^{-1}E$ normalised, $\varphi_\gamma = E_\gamma/\|E_\gamma\|$, which is a polynomial in K applied to E and is computed by the Krylov series (Theorems **??**, **??** and **??**(c)). For it, every inequality in the floor is an equality in its signed form. A roughness function fixed by the inputs alone, such as the cluster seminorm or the manifold high-pass, is a local complexity function: it makes the roughness of the labels computable before training and gives a weaker bound. In practice the only roughness function worth using is therefore the Krylov functional of the current state, and every statement below is made for it first; the fixed-seminorm statements are its corollaries. An earlier draft fixed a Laplacian high-pass throughout, measured its Rayleigh-quotient closure to fail along training, and replaced the closure by a model; that failure is an artefact of the fixed projector, as §1.4 explains, and the model is gone.

Throughout, $R(u) = \|Pu\|$ is a projector seminorm, with P a symmetric projector on sampled functions, so that $C_R = 1$ and $\sigma_R(\theta)^2 = \lambda_{\max}\big(PK(\theta)P\big)$. The rank-one functional is of this form with $P = \varphi\varphi^\top$ for a unit sampled function φ, for which $\sigma_R = \|\varphi\|_K$. Write

$$E_r = PE, \qquad E_s = (I - P)E, \qquad \sigma(t) = \sigma_R(\theta(t)), \tag{1}$$

for the rough and the smooth part of the error under R and for the peak feature roughness along the path. Since Y is fixed, $\dot{E}_r = P\dot{E}$, and $\|E_r(0)\| = R(E_0) \ge R(Y) - R(\hat{Y}_0)$.

1.1 The rough part loses energy only through σ_R

Proposition 1 (Energy of the rough part)**.** *Let N_θ be continuously differentiable in θ and let $\theta(t)$ be the gradient-flow path. Then, with the notation of Eq.* (1)*:*

(a) Exactly,

$$\frac{d}{dt}\,\|E_r\|^2 = -2\,\langle E_r, KE_r\rangle - 2\,\langle E_r, KE_s\rangle, \qquad 0 \le \langle E_r, KE_r\rangle \le \sigma^2\,\|E_r\|^2, \qquad |\langle E_r, KE_s\rangle| \le \sigma\,\lambda_1^{1/2}\,\|E_r\|\,\| \tag{2}$$

(b) For a unit sampled function φ and $P = \varphi\varphi^\top$, the first inequality in (a) is an equality, $\langle E_r, KE_r\rangle = \sigma^2\|E_r\|^2$ with $\sigma = \|\varphi\|_K$, and the signed component obeys, exactly and for every update rule,

$$\frac{d}{dt}\langle \varphi, E\rangle = -\langle \varphi, KE\rangle, \qquad \Big|\frac{d}{dt}\langle \varphi, E\rangle\Big| \le \|\varphi\|_K\,|\dot\theta| = \sigma\,|\dot\theta|. \tag{3}$$

(c) The rate at which the rough part can lose energy is bounded by the peak feature roughness alone once the smooth part is gone: $-\frac{d}{dt}\|E_r\|^2 \le 2\sigma^2\|E_r\|^2 + 2\sigma\lambda_1^{1/2}\|E_r\|\,\|E_s\|$.

Proof. (a) $\frac{d}{dt}\frac{1}{2}\|E_r\|^2 = \langle PE, P\dot E\rangle = \langle E_r, \dot E\rangle = -\langle E_r, KE\rangle$ by Theorem 7(a), and $E = E_r + E_s$. K is positive semidefinite, so $\langle E_r, KE_r\rangle \ge 0$. For a unit ψ in the range of P, $R(\psi) = \|\psi\| = 1$, so $\|\psi\|_K \le \sigma_R$ by the definition of σ_R; hence $\langle E_r, KE_r\rangle = \|E_r\|_K^2 \le \sigma^2\|E_r\|^2$. For the cross term, $\langle a, Kb\rangle = g_a \cdot g_b$, so $|\langle E_r, KE_s\rangle| \le \|E_r\|_K\|E_s\|_K \le \sigma\|E_r\| \cdot \lambda_1^{1/2}\|E_s\|$. (b) $E_r = \langle \varphi, E\rangle\varphi$ and PKP has the single nonzero eigenvalue $\langle \varphi, K\varphi\rangle = \|\varphi\|_K^2$. The rate identity is $\dot E = \Psi\dot\theta$ paired with φ, $\langle \varphi, \Psi\dot\theta\rangle = g_\varphi \cdot \dot\theta$ with $|g_\varphi| = \|\varphi\|_K$ (Proposition **??**(a)); under gradient flow $\dot\theta = -g_E$ gives the first form. (c) is (a) with the two bounds. □

Part (a) is the energy identity of Theorem 7(a) projected on the rough subspace. For a general projector it has two terms, the Rayleigh quotient on the rough subspace, capped by σ^2, and the coupling to the smooth part, capped by $\sigma\lambda_1^{1/2}$. Part (b) is what the Krylov functional changes. The cap on the Rayleigh quotient is attained, so no closure constant appears; and the component $\langle \varphi, E\rangle$ obeys a rate bound with no coupling term at all, because a rank-one seminorm measures a signed component and the sign carries the information that the energy form throws away. The coupling term of (a) is not small for this functional: on the reference problem it is 4 to 12 times the diagonal term from $t = 0$ on [n=200], so an energy bound for $\|E_r\|^2$ built from (a) is empty for it, while the signed bound of (b) is exact. Everything below therefore integrates Eq. (3), not Eq. (2).

1.2 The a priori floor on the rough part

Integrated along the path, Eq. (3) charges the component of the error along φ by the cost of that direction, $\mathrm{cost}_t(\varphi) = \int_0^t \|\varphi\|_{K(s)}|\dot\theta(s)|\,ds$ (Theorem **??**). The a priori form replaces the kernel along the path by its value at the start plus the drift, which Lemma **??** bounds by the second derivative of the network over the ball the path stays in: with $\ell(t) = \int_0^t |\dot\theta|$ the path length and $\bar H_2(\ell)$ the largest second-derivative size over the ball of radius ℓ about θ_0, $\|\varphi\|_{K(s)} \le \|\varphi\|_{K_0} + \bar H_2(\ell)\,\ell(s)$ for every unit φ (Proposition 15(b) in its path form).

Theorem 2 (The rough part is held: a priori form)**.** *Let N_θ be twice continuously differentiable in θ and let $\theta(t)$ be the gradient-flow path, with $\ell = \ell(t)$. Then*

$$\|E(t)\| \;\geq\; F_0(\ell) - \tfrac{1}{2}\,\bar{H}_2(\ell)\,\ell^2, \qquad F_0(\ell) = \max_{\|\varphi\|\leq 1} \Big(\langle \varphi, E_0 \rangle - \ell\,\|\varphi\|_{K_0} \Big), \tag{4}$$

where $F_0(\ell)$ is the smallest error of the first-order expansion about θ_0 of a step of length ℓ, attained by the Krylov functional of (K_0, E_0) at the rate γ with $\Upsilon_0(\gamma) = \ell^2$, and computed by the Krylov series (Theorems **??** *and* **??***; this is Theorem* **??***). For every projector seminorm R with $\sigma_0 = \sigma_R(\theta_0)$,*

$$\|E(t)\| \;\geq\; R(Y) - R(\hat{Y}_0) - \ell\,\sigma_0 - \tfrac{1}{2}\,\bar{H}_2(\ell)\,\ell^2, \tag{5}$$

and under gradient flow $\ell(t) \leq \sqrt{t/2}\,\|E_0\|$ (Proposition **??***).*

Proof. For a unit φ, Eq. (3) integrated gives $\langle \varphi, E(t) \rangle \geq \langle \varphi, E_0 \rangle - \mathrm{cost}_t(\varphi)$, and $\|E(t)\| \geq \langle \varphi, E(t) \rangle$. Insert the drift bound, $\mathrm{cost}_t(\varphi) \leq \ell \|\varphi\|_{K_0} + \bar{H}_2(\ell) \int_0^t \ell(s)\,d\ell(s) = \ell\|\varphi\|_{K_0} + \frac{1}{2}\bar{H}_2(\ell)\ell^2$, and take the maximum over φ. For Eq. (5), take φ the unit vector in the range of P that attains $R(E_0) = \langle \varphi, E_0 \rangle$, for which $\|\varphi\|_{K_0} \leq \sigma_0$, and use $R(E_0) \geq R(Y) - R(\hat{Y}_0)$. The path length bound is Cauchy–Schwarz on $\ell = \int \|E\|_K$ with the energy identity. □

Three features distinguish the theorem. It holds at every time with no horizon and no analyticity, since nothing is expanded. Its first term is exact in the roughness function: $F_0(\ell)$ is the supremum of the fixed-seminorm floors Eq. (5) over all R, so no seminorm has to be chosen and none does better. And the network enters only through K_0, in F_0, and through $\bar{H}_2$: the kernel at the start says what a path of length ℓ can remove to first order, and the second derivative says how much more it can remove by moving the kernel. The earlier draft stated this floor for a fixed projector in energy form, with σ bounded by $\sigma_0 + as$ and the decay charged exponentially; that form carries the coupling term of Proposition 1(a) and is empty for the functional that matters, so it is withdrawn in favour of Eq. (4).

Corollary 3 (Minimum path length and training time to remove the rough part)**.** *Under the hypotheses of Theorem 2, let $\ell^\star_\epsilon = \inf\{\ell \geq 0 : F_0(\ell) - \frac{1}{2}\bar{H}_2(\ell)\ell^2 \leq \epsilon\}$. If $\|E(t)\| \leq \epsilon < \|E_0\|$, then $\ell(t) \geq \ell^\star_\epsilon$ and*

$$t \;\geq\; \frac{2\,(\ell^\star_\epsilon)^2}{\|E_0\|^2 - \epsilon^2}. \tag{6}$$

For every projector seminorm, $\ell^\star_\epsilon \geq \ell^R_\epsilon$, the smallest ℓ with $\ell\sigma_0 + \frac{1}{2}\bar{H}_2(\ell)\ell^2 \geq R(Y) - R(\hat{Y}_0) - \epsilon$.

Proof. Corollary **??**: the right side of Eq. (4) is nonincreasing in ℓ, and $\ell(t)^2 \leq \frac{t}{2}(\|E_0\|^2 - \epsilon^2)$ at the first time the error reaches ϵ. □

Eq. (6) is the a priori ignition time of the rough part of the labels: the path must be long enough for the first-order term to pay for the roughness the labels demand, or the second-derivative term must have moved the kernel, and under gradient flow the path length grows at most as $\sqrt{t}$. The Krylov form $\ell_\epsilon^\star$ is never worse than any seminorm's ℓ_ϵ^R.

1.3 The roughness spectrum and the moving cutoff

The floor of Theorem 2 is one number. Read across the rate γ of the slow part, it locates where the held error sits. For a sampled function u write $u_\gamma = \gamma(K_0 + \gamma I)^{-1}u$ for its slow part under the initial kernel; $\gamma \mapsto \|u_\gamma\|$ is nonincreasing, equal to $\|u\|$ as $\gamma \to \infty$ and to the part of u outside the range of K_0 as $\gamma \to 0$.

$$\rho(\gamma) = \|E_{0,\gamma}\| \tag{7}$$

is the roughness spectrum of the initial error in the coordinates of the kernel: the energy of E_0 that the initial kernel removes at rate γ or slower. For a target with $\hat{Y}_0$ small it is the roughness spectrum of the labels, $\|Y_\gamma\|$, the dataset as the kernel sees it (Part I, the dataset in the coordinates of the kernel); it is fixed by the dataset and the initial kernel and computed by the Krylov series.

Definition 4 (Cutoff rate)**.** *For $t \geq 0$, the cutoff rate is*

$$\gamma^\star(t) = \sup\Big\{\gamma > 0 : \ell(t)\,\tfrac{\sqrt{\gamma}}{2} + \tfrac{1}{2}\,\bar{H}_2(\ell(t))\,\ell(t)^2 \leq \tfrac{1}{2}\,\rho(\gamma)\Big\}, \tag{8}$$

the fastest rate at which, at time t, the a priori bound allows the slow part of the initial error to have lost at most half its size.

Since ρ is nonincreasing in γ and ℓ nondecreasing in t, $\gamma^\star(t)$ is nonincreasing in t: the cutoff moves down through the rates as training proceeds, and it moves even when $\sigma_R(\theta_0) = 0$ on a direction, through the $\bar{H}_2\ell^2$ term, which is feature learning.

Proposition 5 (The error at time t is at least the energy above the cutoff)**.** *Under the hypotheses of Theorem 2, for every t and every $\gamma > 0$,*

$$\|E(t)\| \;\geq\; \rho(\gamma) - \ell(t)\,\tfrac{\sqrt{\gamma}}{2} - \tfrac{1}{2}\,\bar{H}_2(\ell(t))\,\ell(t)^2, \qquad \textit{so} \qquad \|E(t)\| \geq \tfrac{1}{2}\,\rho(\gamma^\star(t)). \tag{9}$$

Proof. Take $\varphi = \varphi_\gamma = E_{0,\gamma}/\|E_{0,\gamma}\|$ in the proof of Theorem 2. In the eigenbasis of K_0 with weights $w_k = \gamma/(\lambda_k + \gamma)$, $\langle \varphi_\gamma, E_0\rangle = \sum_k w_k \hat{e}_k^2/\|E_{0,\gamma}\| \geq \sum_k w_k^2 \hat{e}_k^2/\|E_{0,\gamma}\| = \rho(\gamma)$, and $\|\varphi_\gamma\|_{K_0}^2 = \sum_k \lambda_k w_k^2 \hat{e}_k^2 / \sum_k w_k^2 \hat{e}_k^2 \leq \max_\lambda \lambda\gamma^2/(\lambda+\gamma)^2 = \gamma/4$. □

In words: at time t, the error is at least the part of the initial error that the initial kernel removes more slowly than the cutoff rate, and the cutoff is set by the path length and the roughness capacity of the network, $\sqrt{\gamma}/2$ per unit length along the slow part at rate γ and $\bar{H}_2$ for the motion of the kernel, and by nothing else. Without feature learning

($\bar{H}_2 = 0$) the cutoff is $\gamma^\star$ with $\ell\sqrt{\gamma^\star} \approx \rho(\gamma^\star)$, which is the spectral bias of the lazy regime stated in the kernel's own coordinates; with feature learning the cutoff falls faster, and Corollary 3 says how much faster it can fall at most. For labels whose roughness spectrum ρ has a step at some rate γ_Y, the error is held near $\rho(\gamma_Y)$ until $\gamma^\star(t)$ passes below γ_Y. The chain section replaces the a priori charge $\ell\sqrt{\gamma}/2 + \frac{1}{2}\bar{H}_2\ell^2$ by the actual lifted spectrum along the run: its cutoff bound is this proposition with the weight $\gamma/(\lambda+\gamma)$ hardened to the indicator of $\lambda_k(T)T < 1$ and the slow set transported along the eigenvector flow.

1.4 The closure, revisited

The statements above bound the rough part from below. An earlier draft tried to close them from above with the hypothesis that the kernel's Rayleigh quotient on the rough residual stays a fixed fraction c of σ_R^2, $\langle E_r, KE_r\rangle \geq c\,\sigma_R^2\|E_r\|^2$, measured it with a fixed Laplacian high-pass, found c to decay from 0.02–0.09 to 0.001–0.005 along training while σ_R^2 was lifted 6–50× [n=200], and replaced the closure by a reduced model. That measurement is an artefact of holding the projector fixed: the lift raises the top of the fixed rough subspace while the residual moves to the slowest directions of that subspace, so $c(t) \approx \lambda_{\text{eff}}(t)/\sigma_R(t)^2 \to 0$. For the Krylov functional of the current state, $c = 1$ identically (Proposition 1(b)), and the closure hypothesis is not needed. What is genuinely open is not a closure but the cost: the floor along the path charges every direction by $\int \|\varphi\|_{K(s)}|\dot{\theta}|$, and along the rough directions $\|\varphi\|_{K(s)}$ is lifted by orders of magnitude, which is feature learning; the a priori bound on that lift through $\bar{H}_2$ is a worst case. The two-sided statement that replaces the band is the cutoff bound of Section 2: the energy of the initial error on the transported slow set, with the actual lifted spectrum in place of the a priori charge and the cross-front rotation as the named remainder, and the onset of the lift given by a convergent expansion in the richness parameter.

1.5 Depth and the comparison of architectures

Everything above enters an architecture through $F_0(\ell)$, one Krylov series on the initial kernel and error per design, and through $\bar{H}_2$. For two designs on the same dataset, the comparison principle of Corollary 6 applies with the bounds of this section.

Corollary 6 (Comparing designs on the rough part)**.** *Let designs A and B be trained by gradient flow on the same dataset and let $T > 0$.*

(i) (A priori.) $\|E^A(T)\|$ is at least the right side of Eq. (4) *computed with F_0^A and $\bar{H}_2^A$ at $\ell = \sqrt{T/2}\,\|E_0^A\|$, and at least $\frac{1}{2}\rho^A(\gamma_A^\star(T))$.*

(ii) (Along the run of B, Section 2.) $\|E^B(T)\|^2$ is at most the ceiling of the cutoff bound computed for B: the energy of its initial error on the modes that are still unfit at T under its lifted spectrum, plus the cross-front leakage and the decayed fit part. Its

inputs (the transported unfit set, the leakage, the fit residual) are read along B's trajectory; it is not computable before training.

(iii) If the bound in (ii) is below the bound in (i), then $\|E^B(T)\| < \|E^A(T)\|$: B has learned more of the rough part of the labels by time T. Since (ii) needs B's run, the corollary compares a trained B against an untrained A; a before-training comparison of two designs needs (ii) replaced by an a priori ceiling, which the theory does not have.

The corollary separates what depth does into two effects that can be checked one at a time. A deeper network can have a larger initial kernel on the slow part of the labels, a smaller $\rho(\gamma)$ at every γ; that lowers the held rough error after the lazy phase, since the cutoff of Definition 4 starts lower. And it can have a larger $\bar{H}_2$, the rate at which it can move its kernel per unit of path; that shortens the ignition time of Corollary 3 and advances the onset of the lift given by the richness expansion of Section 2. Both are computed before training, one Krylov series and one nested automatic differentiation per design; the theory says which part of the error each one removes and in what time, but not that either grows with depth, which is a measurement or a statement about the initialization.

1.6 Scope and what to measure

Exact under gradient flow, no hypothesis beyond smoothness. Proposition 1; Theorem 2 and Corollary 3 with $\bar{H}_2$ taken over the ball of radius $\ell(t)$; Proposition 5; Corollary 6(i). **Proved in Section 2, with named inputs read along the run.** The cutoff bound (two-sided, with the leakage as remainder, the leakage not yet evaluated at any sample size); Corollary 6(ii)–(iii) as a trajectory bound; the onset law inside its radius. Nothing in this section is approximated; the closure hypothesis of the earlier draft is not needed, and its failure was the failure of a fixed projector.

What the a priori floor is worth along a run [n=200, 36 runs, to be redone]. Evaluated on the saved kernels and residuals of the ground-truth runs with the Krylov functionals of (K_0, E_0): Eq. (4) is halved by its $\bar{H}_2$ term at a path length of about 0.6, which gradient flow reaches at $t \approx 1$ of 3000, and is negative from $t \approx 10$, because $\bar{H}_2$ is the worst-case curvature and the realised drift of the kernel is far smaller; the first-order term $F_0(\ell)$ alone exceeds the measured error at every later time, by $2\times$ at $t = 100$, $4\times$ at $t = 300$ and $17\times$ at T (medians), so the whole content of the late error is in the motion of the kernel. The path floor with the actual kernel along the run, charged by the measured cost of a functional fixed at the start, holds 0.94 of the error at $t = 10$ and is empty from $t \approx 100$, the cost along that functional having been lifted to $30\times$ its first-order value by then and $280\times$ by T: a floor charged by the lift loses content exactly when the lift arrives. The energy form of the earlier draft is empty from the start for these functionals, its coupling term being 10^3 times its main term at $t = 10$. The one-time floor at the current state, $\|E_{1/t}(t)\|$, holds 0.92 of the error at $t = 10$ and 0.70–0.74 from $t = 100$ on,

and is the floor that stays informative. (Numbers: EXPERIMENTS.md, item C1.) The measurements that decide whether the closure is the right one: (13) the cost $\int \|\varphi\|_{K(s)}|\dot{\theta}|$ of the Krylov functionals along each run against its a priori bound $\ell\|\varphi\|_{K_0} + \frac{1}{2}\bar{H}_2\ell^2$, which is the question of how loose $\bar{H}_2$ is; (14) the slow part $\|E_\gamma(t)\|$ at the current state against $\|E(t)\|$, the one-time floor that is exact to first order; (15) $\rho(\gamma)$ for the labels against the measured error's spectrum in the current kernel basis; (16) the cutoff rate $\gamma^\star(t)$ against the rate at which the measured error's spectrum drops, which tests Proposition 5 as an estimate rather than as a bound; (17) $\rho(\gamma)$ and $\bar{H}_2$ as functions of depth at initialization, which are the quantities through which depth enters Corollary 6.

2 Training through time: the chained expansion and the limits of prediction

Proposition 20 says that no series in $\theta - \theta_0$ reaches the stretch of training in which the rough part of the error is removed, and Section 1 replaced the expansion variable by the peak feature roughness. This section keeps θ as the variable and moves the expansion point instead: training is written as a chain of expansions, each about the current state and each valid within its own radius, whose composition is exact. The chain gives three kinds of statement. Identities that hold at every time for any network in a broad class (§2.1); structure theorems for what gradient flow does to the weights and the kernel at each stage, which is where the low-rank "spikes" and the targeted growth of the kernel come from (§2.3–2.4); and applications in which an open question becomes a computation with the chain or a theorem about its leading orders (§2.5). The last two subsections state why the late stage cannot be predicted from the initialisation without carrying the state forward, in a form that is a theorem rather than an impression (§2.7), and what a statistical description can and cannot promise (§2.8).

2.1 Setting and exact identities

Model class. N_θ is any composition of *parameter-linear blocks* $z_\ell = W_\ell a_\ell$, where a_ℓ is whatever the block receives (dense, convolution, attention projections, normalisation gains), and *parameter-free maps* between them (pointwise activations, softmax, normalisation), differentiable almost everywhere in θ. ReLU qualifies. Training is gradient flow on the squared loss, or any update of the data-indexed outer-product form below (GD, SGD on a subset of the data, momentum through its integral); Adam is treated in §2.5. Write $\Psi(t) = J(\theta(t)) \in \mathbb{R}^{N\times P}$ for the Jacobian on the n data points ($N = n\,n_y$), $K = \Psi\Psi^\top/n$, $\delta_\ell^c(x) = \partial\hat{Y}_c(x)/\partial z_\ell$ for the backprop signal into block ℓ, and $U(t) = \int_0^t E$ for the integrated residual.

Proposition 7 (Four identities)**.** *For every t at which N_θ is differentiable:*

(a) $\dot{E} = -K(t)E$; *hence* $\frac{d}{dt}\|E\|^2 = -2\langle E, KE\rangle$ *and* $E(T) = \mathrm{Texp}\big(-\int_0^T K\big)E(0)$.

(b) $K = \frac{1}{n}\sum_\ell A_\ell \odot D_\ell$, *with* $A_\ell[i,j] = a_\ell(x_i)\cdot a_\ell(x_j) + 1$ *the Gram of the block's inputs and* $D_\ell[(i,c),(j,d)] = \sum_q \delta^c_{\ell,q}(x_i)\,\delta^d_{\ell,q}(x_j)$ *the Gram of its backprop signals.*

(c) $W_\ell(t) - W_\ell(0) = -\frac{1}{n}\int_0^t \sum_i \sum_c \delta^c_\ell(x_i)\,E_c(x_i)\,a_\ell(x_i)^\top$*: the change of every block is a sum of outer products indexed by the data, its row space lies in the span of the block's inputs on the data, and every block is driven by the same residual.*

(d) *For a unit* q *of block* ℓ *with preactivation* z_q *and incoming backward signal* $m_q = W_{\ell+1}^\top \delta_{\ell+1}$,

$$\dot{z}_q = \dot{W}_\ell a_\ell + W_\ell\,\dot{a}_\ell, \qquad \dot{m}_q = \dot{W}_{\ell+1}^\top \delta_{\ell+1} + W_{\ell+1}^\top \dot{\delta}_{\ell+1}.$$

The first terms are given by (c) (the coherent *part); the second carry the motion of the neighbouring blocks through the current weights (the* propagation *part). Together with (a) and (b) this is a closed system on the data.*

Proof. (a) is the chain rule for $\theta' = -\frac{1}{n}J^\top E$. (b): $\partial \hat{Y}_c(x)/\partial W_\ell[q,p] = \delta^c_{\ell,q}(x)\,a_{\ell,p}(x)$. (c): integrate the gradient. (d): differentiate the definitions. □

Remark 8. Everything in this section is a consequence of Proposition 7. In particular the propagation part of (d) is the only place where the initialisation enters other than through the unit statistics; omitting it (as any model built from the kernel alone must) was measured to cost 11–28% of the unit displacements early and a factor 2 in the late error on the reference problem [n=200].

2.2 The stage identity and the chain

Fix stage boundaries $0 = t_0 < t_1 < \cdots < t_S = T$ and write $\delta\Psi_s = \Psi(t_s) - \Psi(t_{s-1})$.

Proposition 9 (Stage identity)**.** $K(t_s) - K(t_{s-1}) = \mathrm{Cross}_s + Q_s$ *with* $Q_s = \delta\Psi_s\delta\Psi_s^\top/n \succeq 0$ *and* $\mathrm{Cross}_s = (\Psi\delta\Psi_s^\top + \delta\Psi_s\Psi^\top)/n$, $\Psi = \Psi(t_{s-1})$. *For a unit vector* v*, the quadratic form changes by* $\frac{1}{n}\big[2\langle\Psi^\top v, \delta\Psi_s^\top v\rangle + \|\delta\Psi_s^\top v\|^2\big]$*, which is positive whenever* $\|\delta\Psi_s^\top v\| > 2\|\Psi^\top v\| = 2\sqrt{n\lambda_v}$.

Theorem 10 (Feature-sum identity and bracket at T)**.** $\Psi_T = \Psi_0 + \sum_s \delta\Psi_s$*, so for every unit* v*, with* $D_v = \sum_s \delta\Psi_s^\top v$ *and* $\ell_v = \|\Psi_0^\top v\|$,

$$\tfrac{1}{n}\big(\|D_v\| - \ell_v\big)_+^2 \;\le\; v^\top K_T v \;\le\; \tfrac{1}{n}\big(\|D_v\| + \ell_v\big)^2.$$

D_v *splits exactly into coherent and propagation parts by Proposition 7(d). On the directions the residual has driven,* ℓ_v *is negligible and the lower bound is the value: measured* 0.90–1.00 *of* $v^\top K_T v$ *for every rank* ≥ 60 *on the reference problem [n=200]. Where* J *is* C^1 *along the path,* $|\lambda_k(K_T) - \lambda_k(K_0)| \le \int_0^T C(s)\|E(s)\|\,ds$ *with* $C = \frac{2}{n^2}\|\Psi\|_2^2\|\Psi'\|_2$ *(Weyl), so every lifted eigenvalue at* T *is bracketed between two functionals of the residual history.*

Theorem 11 (Stage error and the chain)**.** *Where J is C^2 along the path, $\delta\Psi_s = \alpha_s\dot\Psi(t_{s-1}) + R_s$ with $\|R_s\| \le \frac{\alpha_s^2}{2}\sup\|\ddot\Psi\|$; choosing $\alpha_s\|\ddot\Psi\| \le \varepsilon\|\dot\Psi\|$ makes each stage's first-order increment accurate to relative ε, and the composition of the stages is the exact trajectory. The quantity $\|\ddot\Psi\|/\|\dot\Psi\|$ is the inverse of the stage length over which the linearisation at the current state is trustworthy; its norm along the descent direction is H_2.*

2.3 What a stage does to the weights: the spike theorem

For a hidden block whose outgoing weights are $W_{\ell+1}$, write $S[i,q] = \sigma'(z_q(x_i))$ for the local derivative of the map following the block, $M[i,q]$ for the backprop coefficient contracted with the drive (for the last hidden block $M = U\,W_{\text{out}}$, of rank $\le n_y$), and A for the block's inputs on the data. By Proposition 7(c), $\Delta W = -\frac{1}{n}(S \odot M)^\top A$.

Theorem 12 (Core and bulk)**.** *(a) (*at initialisation*) Split $S = h_0 + (S - h_0)$ with h_0 the mean of σ' over the data. Then $\Delta W = \Delta W^{\text{core}} + \Delta W^{\text{bulk}}$ with $\Delta W^{\text{core}} = -\frac{h_0}{n}M^\top A$ of rank $\le n_y$, its left singular space spanned by the rows of W_{out} and its right singular space by $A^\top U$. If the block's input weights are independent of $W_{\ell+1}$ (true at θ_0), the expected bulk-to-core energy ratio is $\rho = \mathrm{Var}(\sigma'(z))/\mathbb{E}[\sigma'(z)]^2$ at the preactivation scale. The output block has rank $\le n_y$ and the input block rank $\le d_{\text{in}} + 1$ exactly.*

*(b) (*any time, no independence*) For any rank-r approximation S_r of S with remainder $\tilde S$, $\mathrm{rank}\big((S_r \odot M)^\top A\big) \le n_y r$ and $\|(\tilde S \odot M)^\top A\|_F \le \|\tilde S\|_{\max}\|M\|_F\|A\|_2/n$. Since $Z = A\,W_\ell^\top + b$, the numerical rank of $S = \sigma'(Z)$ is controlled by the spectrum of the block's input Gram and by the Hermite coefficients of σ' at the current preactivation scale, both quantities the chain carries; for deeper blocks the bound composes multiplicatively down the stack.*

Proof. (a) Expand σ'; the core is $h_0 M$; the bulk's expected energy factorises by independence of S_{iq} and M_{iq}. (b) $\mathrm{rank}(X \odot Y) \le \mathrm{rank}X \cdot \mathrm{rank}Y$, the entrywise Frobenius bound, the column-space inclusion for Z, and the Hermite expansion. □

2.4 The richness expansion, its radius, and the chain as its resummation

Let ϵ be any parameter that multiplies the weight velocity at fixed residual dynamics in a suitable time: the inverse output scale $1/s$ with $\tau = s^2 t$ for centred outputs $\hat Y = s(N_\theta - N_{\theta_0})$; the Euler step Δ of §2.5; a μP-type multiplier; an inverse-width factor. Then

$$\frac{dE}{d\tau} = -K(\theta)E, \qquad \frac{d\theta}{d\tau} = -\frac{\epsilon}{n}J(\theta)^\top E.$$

Theorem 13 (All-orders richness expansion)**.** *For analytic parameter-free maps and each fixed τ there is $R(\tau) > 0$ such that every analytic observable of the state is a convergent*

power series in ϵ for $|\epsilon| < R(\tau)$; in particular

$$\frac{v^\top K(\tau) v}{v^\top K(0) v} - 1 = \sum_{k \geq 1} a_k(\tau)\, \epsilon^k,$$

with coefficients determined recursively from θ_0 and the lazy trajectory alone: a_0 is kernel regression with $K(\theta_0)$, $\theta^{(1)}(\tau) = -\frac{1}{n} J_0^\top U_{\text{lazy}}(\tau)$, $a_1 = 2\langle \Psi_0^\top v, J'[\theta^{(1)}]^\top v \rangle / (n\lambda_v)$, and a_k involves derivatives of J at θ_0 up to order k along lower-order displacements and the lower-order residual corrections (linear inhomogeneous equations, variation of constants in the eigenbasis of K_0). No quantity along a trained path enters at any order.

Theorem 14 (Radius, and the chain as resummation)**.** *$R(\tau) > 0$ by analyticity. Within the region in which the stage criterion of Theorem 11 holds at θ_0 the first-order term of the series is accurate to the stated relative error, and along a direction v the first-order lift is a valid approximation while $\|\delta\Psi^\top v\| < \|\Psi_0^\top v\|$ (Proposition 9); these are truncation statements and do not bound $R(\tau)$ from below, which would need a majorant estimate on the derivatives of J at θ_0 (available for analytic activations on a ball; not carried out). Where the lift along v reaches order one the truncated series fails sharply in the measurements, consistent with the singularity of $\epsilon \mapsto$ trajectory sitting there. Re-expanding at each stage boundary (Theorem 11) continues the series past $R(\tau)$ by moving the expansion point along the path; within $R(\tau)$ the two agree, beyond it only the chain is defined.*

Proof. Analytic dependence of ODE solutions on parameters; the recursion is the Taylor expansion of the vector field in ϵ; the truncation statements are Proposition 20 at the expansion point θ_0 together with Proposition 9; the resummation uses only $R(\tau) > 0$. □

2.5 Applications

Euler versus direct. Let $y_D = \Phi_\Delta(x)$ and $y_E = (\Phi_\Delta(x) - x)/\Delta$ be the direct and the Euler target, trained from the same θ_0 with centred outputs, and write the direct residual on the increment as $E_D = \Delta \tilde{E}$.

Theorem 15. *$\tilde{E}$ and E_E obey the same residual equation with their own kernels, start from the same value, and the weight velocities satisfy $\dot{\theta}_D = -\frac{\Delta}{n} J^\top \tilde{E}$ against $\dot{\theta}_E = -\frac{1}{n} J^\top E_E$: the direct problem is the Euler problem with the richness parameter multiplied by Δ (Theorem 13 evaluated at a smaller argument). Hence (i) direct's stage lifts on the increment are Δ^2 times Euler's at equal normalised residual, and (ii) as $\Delta \to 0$ at fixed budget the direct network is in the lazy regime on the increment, after a Δ-independent early phase in which it fits the identity component of its target, and its normalised error converges to the frozen-kernel error of the increment target.*

For direct as $\Delta \to 0$ the held-rough statements of Section 1 are therefore frozen-kernel statements (no lift, fixed basis); they are not tight, since the floor decays at the rate set by

the top of the rough subspace while the frozen-kernel error decays mode by mode. Euler's error is the full feature-learning error, below the frozen-kernel value by the feature-learning gain, which grows with the budget and with the increment's roughness; this comparison is conditional on the rate comparison below. The static argument of the previous sections ("the increment is a smoother target") is the frozen-kernel shadow of this: dividing by Δ multiplies the drive by $1/\Delta$ and the lifts by $1/\Delta^2$, so Euler trains rich on the increment while direct trains lazy on it.

Progressive sharpening, the initial reduction phase, and the edge of stability. By Proposition 9, $\dot{\lambda}_1 = v_1^\top(\mathrm{Cross} + Q)v_1$ per stage with $Q \succeq 0$. Along the residual the cross term at $t = 0$ equals $-\frac{2\alpha}{n^2}\langle E, N''_\theta[g, g]\rangle$ with $g = J^\top E$, exactly, and it is negative on 90–91% of the eigen-directions for the raw residual $E = \hat{Y}_0 - Y$ and on 48–55% (no systematic sign) for the centred residual $-Y$ [n=200]: the initial sharpness-*reduction* phase reported in the warmup literature is the network unlearning its own initial function and is absent for centred initialisations. Once learning has accumulated, the coherent part of the cross term is the inner product of the accumulated core above with the stage core below through the shared drive, positive while the drive keeps its direction (measured positive on 80–100% of modes from $t \approx 1$ on): progressive sharpening, faster with depth by composition ($5\times, 11\times, 20\times$ top gain at depths $2, 3, 6$ on the $\Delta = 0.5$ target and $86\times$ at depth 6 on $\Delta = 1.0$ [n=200]). Taking GD steps as stages, above the stability threshold the top component of the residual alternates in sign, the stage cores along it alternate, their cross terms with the accumulated core are negative at the core level, and λ_1 decreases until the step is stable again: sharpening and self-stabilisation are the two signs of one term. What is not proved is the sign of the init-feature part of the cross term after the transient (Lemma 16 below).

Adam. For $\dot{\theta} = -P(t)\nabla L$ with P diagonal positive, (a) of Proposition 7 holds with $K_P = \Psi P \Psi^\top/n$, (c) with the outer products multiplied entrywise by P_ℓ, and Theorem 10 at a fixed time with $P_T^{1/2}$-weighted features. Along the path the stage identity acquires a third term: with $\delta P_s = P(t_s) - P(t_{s-1})$, $K_P(t_s) - K_P(t_{s-1}) = \mathrm{Cross}_{P,s} + Q_{P,s} + \Psi(t_s)\,\delta P_s\,\Psi(t_s)^\top/n$, and δP_s is negative on every coordinate whose second-moment estimate is still growing. The lift and the preconditioner's decay therefore compete, the monotone-spectrum inputs below do not transfer to K_P without a bound on this term, and with momentum the drive is the averaged gradient, so the energy identity has no sign. What transfers is the set of identities, the fixed-time bracket and the spike statement; the discrete regime in which Adam runs in practice (its own edge of stability) is outside gradient flow altogether. The spike statement needs one more fact: P is built from the gradient block itself, so $P \odot G$ is a normalised sign pattern of a rank-$\leq n_y$ matrix, and the same mean-plus-fluctuation argument as Theorem 12 gives its core fraction as the squared first Hermite coefficient of the normaliser under the entry distribution of G: $2/\pi$ for Gaussian-like entries,

1 for coordinate-aligned (sparse) singular vectors, with the time-averaged second moment lowering it further (measured 0.68–0.81, 0.95–1.00, 0.52 respectively on synthetic blocks). This is the rotation-dependent property that distinguishes Adam from SGD.

Validation, and minibatches. With J_{val} the Jacobian on validation points, $\dot{\hat{Y}}_{\text{val}} = -K_{vt}E$ with $K_{vt} = J_{\text{val}}J^\top/n$, exactly, so $-\frac{d}{dt}\frac{1}{2}\|E_{\text{val}}\|^2 = E_{\text{val}}^\top K_{vt}(0)E + E_{\text{val}}^\top[K_{vt}(t) - K_{vt}(0)]E$: a lazy-transfer part and a feature-learning-transfer part. The coherent part of the lift on validation points is the training stage core transported by the cross activation Gram; under equal unit statistics on the two samples the whole chain applies with that substitution. The same identity covers minibatch training: a step on a batch B drives the residual on the full set by $\dot{E} = -K_{\text{full},B}\, g_B$, which is the validation identity with B as the training set, so the chain is exact for SGD with the drive history $\{g_{B_s}\}$, while the full-set energy, like the validation energy, has no sign step by step. Measured [n=200]: early both parts positive and the validation rate 70–110% of the training rate; late the lazy part negative (the frozen kernel maps a residual living in sample-specific rough modes into directions that increase validation error, the static held-rough mechanism seen as a sign) while the lifted part stays positive. At the sample sizes of interest, where training and validation agree, the sign pattern is a prediction to be checked rather than a result.

Delayed feature learning ("grokking"). The mechanism is Theorems 12–10: the residual drives a positive lift along the directions it has visited, so a direction too slow for the initial kernel to fit within the budget is fit later through the lift. The onset is Theorem 13 with $\epsilon = 1/s$: an init-computable series, verified to 0–16% in its range, with a proved radius; the delay to a small lift is its inversion, and because a_1 changes sign no universal exponent in s exists, which accounts for the disagreement among reported scalings. The event itself, a lift of order one or more along a misaligned direction, lies outside $R(\tau)$ by construction and is a chain computation; by §2.7 it is not predictable from the initialisation in any other way. Rigorous timings in the literature (weight decay, modular arithmetic, two-layer mean field) are special cases evaluated inside their own radii.

The error at T, term by term. Write $K(t) = K_0 + \Delta(t)$ with Δ the accumulated stage increments. The residual equation has the exact Dyson form

$$E(T) = \sum_{m\geq 0}(-1)^m \int_{T>u_1>\cdots>u_m>0} e^{-K_0(T-u_1)}\Delta(u_1)\, e^{-K_0(u_1-u_2)}\cdots\Delta(u_m)\, e^{-K_0 u_m} E_0,$$

absolutely convergent for bounded Δ (the m-th term is bounded by $(\int\|\Delta\|)^m/m!\,\|E_0\|$), so the series is the solution, not an approximation. Every term is a product of lazy propagators and increments, and every increment is built from the stage cores of Theorem 12, so $E(T)$ lies in the span generated by the lazy flow of E_0 and the stage targets. The first-order term paired with the lazy solution is $-\int_0^T \langle E_{\text{lazy}}(2T-u), \Delta(u)E_{\text{lazy}}(u)\rangle\, du$, whose sign is

fixed by two properties at each u: $\Delta(u) \succeq 0$ as an operator on the subspace spanned by the lazy residual at u and at $2T-u$ (the sign lemma below) and the alignment of $E_{\text{lazy}}(u)$ with $E_{\text{lazy}}(2T-u)$ in the modes Δ touches (persistence; automatic in a fixed basis). Under both, $\|E(T)\|^2 \le \|E_{\text{lazy}}(T)\|^2 - 2\int_0^T \langle E_{\text{lazy}}(2T-u), \Delta(u) E_{\text{lazy}}(u)\rangle du + R_2$, with an explicit nonnegative first correction and a remainder bounded by $\|E_0\|^2(D^2 + 2e^D r + r^2)$, $D = \int_0^T \|\Delta\|$, $r = e^D - 1 - D$, from the Dyson bound on the terms $m \ge 2$. The remainder is not sign-controlled by the two properties once the basis rotates; in a fixed basis the series resums exactly to the mode-wise bound of Section 1, and there the two properties are the only hypotheses to all orders. With rotation the first-order inequality is what the expansion proves, and the all-orders comparison is organised differently below.

What the data say about the two hypotheses. On the reference run every initial eigen-direction carrying the late residual satisfies the feature-change inequality of Theorem 10, so the accumulated increment is positive along each of them [n=200]. This is evidence for the sign condition on that subspace, not a proof of it: positivity of a quadratic form on each of a set of directions does not give positivity on their span, the condition the theorems use is the smallest eigenvalue of the increment compressed to the subspace, and the monotone-spectrum statements below need it for each stage increment, not only for the accumulated one. What fails outright is persistence in the initial basis: the subspace carrying the residual at $t = 44$ is only 55% retained at $t = 428$, the accumulated increment on those modes being 10^2–10^3 times the initial gaps. Persistence is therefore false there, not merely unproved, and the error statement must be organised in the instantaneous eigenframe, where the residual's rank profile moves slowly (the front of Section 1) even though its direction does not; the fixed-basis bounds are then the adiabatic limit and the rotation enters as a named remainder, coupling over current gap, per stage. That derivation is the remaining step.

The moving-frame identity. With $K(t) = V\Lambda V^\top$ and $c = V^\top E$ the residual in the instantaneous eigenframe, $\dot c = -\Lambda c - Ac$ with $A = V^\top \dot V$ antisymmetric, so $\frac{d}{dt}\|c\|^2 = -2c^\top \Lambda c$ exactly: rotation redistributes energy among the current modes and does not change the norm. If every stage increment from t_0 on is positive semidefinite on a subspace containing the residual, then $\langle E, K(t)E\rangle \ge \langle E, K(t_0)E\rangle$ and the descent rate is at least the frozen-kernel rate on the *current* residual; if the increments are positive on the whole space, the sorted eigenvalues are nondecreasing (Weyl) and the same reads, in the moving frame, as the frozen sorted spectrum evaluated on the current rank profile. Neither form needs eigenvector persistence; both need per-stage positivity, which is the sign lemma below stated per stage and is a hypothesis, checked on the data by the smallest eigenvalue of the compressed stage increment. What a closed comparison with the lazy trajectory then needs is the evolution of the rank profile, i.e. the moving cutoff of Section 1: in the moving frame persistence is a statement about where the residual sits, not which way it points,

and the fixed-basis bounds are the case $A = 0$.

The cutoff bound. The slow part of the error at rate γ, $E_\gamma = \gamma(K + \gamma I)^{-1}E$, keeps each mode with the weight $\gamma/(\lambda_k + \gamma)$; replacing the weight by the indicator of $\lambda_k < \gamma$ at $\gamma = 1/T$ gives the following hard-threshold form, in which the energy identity closes exactly. Fix the index set $u = \{k : \lambda_k(T)T < 1\}$ of modes still unfit at T, with cutoff rank $k^* = \min u^c$, and transport it along the eigenvector flow by continuity (through crossings, which are massive here: sorted rank is not an invariant label), $P(t) = V(t)\Pi_u V(t)^\top$. With $E_u = PE$, $E_f = (I - P)E$ and the cross-front rotation rate $\rho = \|A_{uf}\|_2$, the spectral norm of the $u \times f$ block of A,

$$\tfrac{d}{dt}\|E_u\|^2 = -2E_u^\top K E_u + L, \qquad \tfrac{d}{dt}\|E_f\|^2 = -2E_f^\top K E_f - L, \qquad |L| \le 2\rho\|E_u\|\|E_f\|,$$

the leakage $L = 2c_u^\top A_{uf} c_f$ being the only coupling between the two subspaces. Hence, with λ_u, λ_f the extreme eigenvalues of K on the two tracked subspaces at each time,

$$\|\Pi_u c(t_0)\|\, e^{-\int_{t_0}^T \lambda_u} - \int_{t_0}^T \rho\|E_f\| \;\le\; \|E(T)\|, \qquad \|E(T)\|^2 \le \|\Pi_u c(t_0)\|^2 + 2\int_{t_0}^T \rho\|E_u\|\|E_f\| + \|E_f(T)\|^2,$$

with $\|E_f(T)\| \le \|(I - \Pi_u)c(t_0)\|e^{-\int \lambda_f} + \int \rho\|E_u\|$. At $t_0 = 0$, $\Pi_u c(0)$ is the energy of the initial error on the init-time eigenvectors that evolve into the unfit set at T (not the energy above sorted rank k^* at $t = 0$, which mislabels the modes by an order of magnitude here), the cutoff being set by the *lifted* spectrum at T; computing it requires the continuously tracked eigenvectors, which the chain carries. Three things should be said about the inputs. A continuously tracked eigenvalue is not monotone even when the sorted spectrum is, so $\int \lambda_u$ is a run quantity; if the tracked eigenvalues in u are nondecreasing, then $\int_{t_0}^T \lambda_u < 1$ and the floor is at least $e^{-1}\|\Pi_u c(t_0)\|$ minus the leakage, which is the bracket's real content. For a generic path the eigenvalues do not cross but avoid each other, and at an avoided crossing between a mode in u and a mode in f the coupling over gap is large; those events are exactly where ρ is large, so the order-of-magnitude mislabelling seen with sorted ranks does not vanish with continuity transport but moves into the leakage integral, which has not been evaluated at any sample size. And rotation among unfit modes, where the term-wise corollary (iii) below places most of it, moves energy within P and costs nothing, while rotation across the front is weighted by the fit residual, which is small except at the crossing events themselves. No persistence hypothesis appears; the inputs are the tracked cutoff set, the leakage, and per-stage positivity of the increments on the residual's subspace for the decay rates. An equivalent formulation defines the unfit set at each time by the current spectrum, $u(t) = \{k : \lambda_k(t)t < 1\}$, the moving cutoff of Section 1; then P jumps when a mode's $\lambda_k t$ crosses one and the leakage carries those jumps explicitly. The cutoff rank is bracketed two-sidedly by Theorem 10 through the accumulated feature change, ρ is the per-stage coupling over current gap across k^*, and the static held-rough floor of

Section 1 is the case in which the cutoff is read from the frozen spectrum: the lift moves the cutoff up in rank, and that is the whole effect of feature learning on the error, in one scalar. For a direction v, the time it leaves u solves $\lambda_v(T_v)T_v = 1$ and is bracketed by $(\|D_v(t)\| \mp \ell_v)^2 t/n = 1$: the delayed-feature-learning event as an exact crossing condition with a two-sided bracket from the accumulated drive and a sharpness set by ρ at the crossing.

The rate comparison, and the one lemma it rests on.

Lemma 16 (Sign of the cross term; status: derived for the core part, measured for the rest)**.** $v^\top \mathrm{Cross}_s v = \frac{2}{n}\big[\langle \Psi_0^\top v, \delta\Psi_s^\top v\rangle + \sum_{r<s}\langle \delta\Psi_r^\top v, \delta\Psi_s^\top v\rangle\big]$. *The second term's core-core part is nonnegative while the integrated residual keeps its direction (shared drive); the first term is measured negative at $t = 0$ for raw residuals and sign-free for centred ones, positive after the transient, and is not derived.*

If the lemma holds from t_0 on, then $\dot{L}(t) \le -\langle E, K(t_0)E\rangle$ for $t \ge t_0$: feature learning never slows the descent below what the kernel at t_0 would produce on the current residual. A closed-form bound on $\|E(T)\|$ needs in addition a persistence condition on the residual direction, because $K(t) \succeq K(t_0)$ does not control mode-wise decay when the eigenvectors rotate, and they rotate by about half in the relevant band by mid-training [n=200].

As a theoretical object the chain is exact: infinitely many expansion points, each to all orders, is the flow itself, and everything above uses only that. How a truncation at one stage reaches T (the error accounting that makes every reduced version of the chain a controlled approximation), and the numerical verifications of the identities and of the leading orders, are collected in Appendix A.

Term-wise corollaries. Once the right series is used, several statements become unconditional. (i) Direct is the richness series at argument $\Delta\epsilon$, so its k-th lift term is Δ^k times Euler's: the lift comparison holds order by order inside the radius with no sign lemma. (ii) By the variational principle $\lambda_{\max}(K_0+\Delta) \ge \lambda_1 + v_1^\top \Delta v_1$ and $\lambda_{\min}(K_0+\Delta) \le \lambda_{\min} + v_{\min}^\top \Delta v_{\min}$ for any symmetric Δ, so the totality of the higher-order terms pushes the top eigenvalue up and the bottom down, exactly and with no convergence question (the Rayleigh–Schrödinger series itself converges for $\|\Delta\|$ below half the eigenvalue's gap, which the chain's stages satisfy for the top and only at fine resolution for the tail); its second-order term $\sum_{k>1} |v_1^\top \Delta v_k|^2/(\lambda_1 - \lambda_k) \ge 0$ is the leading piece: progressive sharpening of the top mode has an unconditional second-order component, and the deep tail an unconditional second-order depression, which is the measured early decrease of the kernel along init-tail directions before the first-order lift overtakes it. (iii) Mode k rotates by $\sum_{j\ne k}(v_j^\top \Delta v_k)/(\lambda_k - \lambda_j)\, v_j$, coupling over gap; with an init spectrum $\sim k^{-a}$ the relative gap is $\sim a/k$, so the rotation grows with rank, i.e. with roughness: the fixed-basis assumption fails exactly where the roughness theory places the action, and |coupling|/|gap| is a computable criterion for when

a direction keeps its identity. (iv) Two initialisations with the same unit statistics differ in every Dyson term only through the propagation part of the stage features, so the spread of $E(T)$ over initialisations is bounded by the propagator times the accumulated propagation mismatch: the irreducible uncertainty of any statistical description is an explicit, measurable, width-dependent quantity, and whether the statistical chain becomes exact at large width is the question whether that quantity shrinks with width.

2.6 Consequences for the held rough error

The results above replace the closure of Section 1 and sharpen its exact part. (i) Proposition 1 is the energy identity projected on a fixed P; the cutoff bound is the same identity with P transported along the eigenvector flow, the static off-diagonal leak becoming the cross-front rotation, and the fixed-P statements are its $A = 0$ reading. (ii) Proposition 15(b) is the Hellmann–Feynman lift bound; its worst-case integral $\Sigma(s) = \sigma_0 + as$ is replaced by the two-sided trajectory bracket of Theorem 10 on every eigenvalue of PKP. (iii) The efficient-supply closure of the earlier draft must be dropped: along training the Rayleigh quotient of the kernel on the rough residual is not a fixed fraction of σ_R^2 but decays, $c(t) \approx \lambda_{\text{eff}}(t)/\sigma_R(t)^2 \to 0$, because the lift raises the top of the rough subspace while the residual moves to its slowest directions, the front [n=200: $c = 0.02$–0.09 at $t = 0$, 0.001–0.005 late, with σ_R^2 lifted 6–50×]. The two-sided band built on it is replaced by the cutoff bound, whose only remainder is the leakage. (iv) The error is not held in any fixed roughness seminorm: the kernel lifts the Laplacian-rough directions and fits them, the whole residual becomes Laplacian-rough while decaying, and what is held is the energy on the modes still unfit in the *current* basis. Roughness must be measured against the lifted kernel, which is what the moving frame does. (v) The a priori cutoff of Definition 4 becomes the transported cutoff read from the lifted spectrum; Proposition 5 remains a valid a priori floor. (vi) Proposition 20 is the radius theorem at a single expansion point and the chain is its continuation. (vii) The reduced model of ignition is subsumed by the richness expansion and the crossing condition and can be cut. (viii) The roughness function in all of the above is the one of Part I: the Krylov functional $\varphi_\gamma = E_\gamma/\|E_\gamma\|$ built from the current error and the current kernel, which attains the supremum of the floor over all seminorms. The transported projector of the cutoff bound is its hard-threshold form, the unfit set $\{\lambda_k(t)t < 1\}$ standing in for the weight $\gamma/(\lambda+\gamma)$ at $\gamma \approx 1/t$, and the cutoff bound is the energy identity for that slow part; no input-space notion of roughness enters, and the fixed-projector statements of Section 1 are the weaker corollaries one obtains by holding a seminorm fixed.

2.7 Why the late stage is not predictable from the initialisation

Proposition 17 (Determinism; no chaos)**.** *For analytic N_θ the gradient-flow trajectory is real-analytic in t, so the history on any interval determines $E(T)$. The loss is a strict*

Lyapunov function, so there are no recurrent sets and no sustained exponential separation of nearby trajectories; what remains is finite-time stretching along transiently negative-curvature directions and algebraic amplification through the lift. [n=200: over the full run a 10^{-7} *relative perturbation of* θ_0 *grows by a factor* 6 *(random direction) and* 21 *(tail-feature direction), while a top-feature perturbation is contracted* $16\times$ *during the fit of the top mode and regrows to* $5\times$ *through the lift; the induced residual perturbation grows by two to three orders of magnitude and ends at* 10^{-6} *relative. The growth is algebraic throughout, roughly* $t^{0.2\text{–}0.4}$ *after* $t \approx 10$*, with no exponential phase in* 3.5 *decades of time.]*

So the obstruction is conditioning, and it has three parts, each a theorem.

Theorem 18 (Hidden state)**.** *Let* A *and* Δ *be the input activations and the outgoing backprop signals of a hidden-hidden block on the data at* θ_0*. For every* $N = UV^\top$ *with* $U^\top\Delta = 0$ *and* $V^\top A = 0$ *(such* N *exist whenever the widths exceed the ranks, e.g.* $H > n\,n_y$*, and have any norm), the initialisation* $\theta_0 + N$ *has the same unit states, output, residual, loss, gradient and kernel on the data, and differs only in the propagation part of Proposition 7(d). Hence* $E(T)$ *is not a function of* (K_0, E_0, Y)*, nor of any finite number of their time derivatives at* 0 *when the widths allow the corresponding null spaces. [Verified: with* $\|N\|_F = \|W\|_F$ *the states agree to* $5 \cdot 10^{-15}$ *and the kernel increments after one step differ by* 1%*.]*

Proposition 19 (Visibility)**.** *Let* B *be the modes of* $K(t_0)$ *carrying a fraction* ϕ *of* $\|E(T)\|^2$*. Their share of the loss-decrease rate at* t_0 *is at most* $\max_B \lambda_k(t_0)/\lambda_{\text{eff}}(t_0)$*, and in the fixed-basis solution their components are constant on* $[0, t_0]$ *to relative order* $\lambda_k(t_0)t_0$*. [n=200: median* $1.4 \cdot 10^{-6}$ *at* $t_0 = 0.01T$ *and* 10^{-2} *at* $t_0 = 0.1T$*; share of the rate* $8 \cdot 10^{-7}$ *and* $7 \cdot 10^{-4}$*.]*

Proposition 20 (Amplification)**.** *The same modes are later lifted by the gain of Theorem 10 (measured* 10^2–10^4*), coherently across blocks, quadratically in the accumulated drive, and* $E_k(T) \sim \exp(-\int \lambda_k)$ *is exponentially sensitive to the lifted eigenvalue. In the fixed-basis model the sensitivity of* $\log|E_k(T)|$ *to a mode's growth exponent is* $2\Lambda_k(T)\log(T/t_0)$ *with* $\Lambda_k(T) \approx 1$ *for the modes that matter.*

Corollary 21 (The conditioning statement)**.** *To predict* $E(T)$ *to relative precision* ε *from the history up to* t_0 *the tail components must be known to relative precision of order* $\varepsilon\,\lambda_k(t_0)t_0$ *divided by the amplification, i.e. about* 10^{-7} *at* $t_0 = 0.01T$ *for* $\varepsilon = 0.1$ *[n=200]: five orders of magnitude below the initialisation-to-initialisation variation of the early history (*$2\cdot10^{-2}$*) and below single precision. The late state is a smooth, deterministic function of the early state whose relevant derivatives are enormous exactly in the directions that are unobservable early.*

The three parts have a common root: the drive is the residual, the residual lives in the slow modes of the current kernel, and the slow modes are both the ones that have not

moved yet (invisible) and the ones feature learning targets (amplified). Feature learning is the mechanism by which the network amplifies precisely the information that no early observable contains, and that information is seeded by the initialisation through Theorem 18. This is why every early predictor we tried (frozen kernel, gain-law extrapolation, residual front, the loss itself) orders designs within 1.1× at chance and within 1.3× at 65–76%, why seed pairs that agree to 2% for the first tenth of training end 15% apart, and why the only object that reproduces the late error is the trajectory carried forward [n=200].

2.8 What a statistical description can promise

The impossibility above is a statement about one network. A statistical description does not predict one network; it predicts the distribution over initialisations, and by Theorem 18 that is the correct object, because the quantity that defeats prediction is the draw of the hidden state. Three consequences follow, two of them proved and one to be measured.

The irreducible uncertainty of any statistical prediction for a single run is the spread of the late error over initialisations, 15% median here [n=200]. A design comparison finer than that spread is not well posed for one run of each design, whatever the method; it is well posed for the expected error, which is a statistic.

The coherent chain is the statistical approximation: replace the network's own units by draws from the unit distribution and the propagation part by its average, and Propositions 7(b,c) and Theorem 12(a) close on the data. On the one case run to convergence [H=96, depth 2, n=200] it reproduced the structure, the gain law and the ordering of designs, with a 2× bias in the late error and a 4–5× excess in the tail gain traced to the averaged propagation term and to the transfer of the ensemble's lift; at depth 3 the bias was larger (5× in the tail gain). It was not run at larger width. Whether the bias and the seed spread decrease with width, as the propagation term being a finite-width fluctuation suggests, or persist with depth, as the depth-3 result warns, is the single most consequential unmeasured quantity in this work, and it decides the question in the next paragraph.

The diagnostic that is well posed is the expected late-stage ability of an architecture. If the spread over initialisations shrinks with width and the coherent chain converges to the exact one in that limit, then ranking architectures by their expected late error is both meaningful and computable without the particular initialisation, and the finite-width exponent of the convergence is the error bar. Everything in this section is set up for that measurement: the exact chain for the per-initialisation truth, the coherent chain for the expectation, and the seed spread for the floor. The ranker built from the identities (lazy error, H_2, top eigenvalue, output scale) reached a 1.3× resolution out of sample [n=200], which is the expected accuracy of a one-Jacobian statistic and should be read as a lower bound on what the expected-error ranking can do.

2.9 Scope and what to measure

Every number in this section comes from $n = 200$ training points on the Lorenz-63 one-step map, at which the training set is nearly fit within the budget, the residual-adapted late phase is short, the depth-6–8 "floor" is a sample-size artefact, and the negative lazy-transfer sign in the validation split is overfitting of a 200-point sample. The theorems carry n as a symbol and are unaffected; the verifications, bulk ratios, exponents, signs, transfer fractions and the ranker's fits are all $n = 200$ statements and are to be redone at a sample size that leaves a genuine late stage before any is quoted. The exact chain does not scale to large n (one $N \times N$ eigendecomposition per stage); the ground-truth runs do, so the redo is training-only, and it should include, at that n: the seed spread of the late error against width and depth, with depth one as the control in which the hidden-state channel of Theorem 18 is absent; the gap between the own-initialisation and the other-initialisation chain against width and depth; the per-stage sign of the increments as the smallest eigenvalue of their compression to the residual's subspace; the spike bound along the trajectory; the transported cutoff bracket with its leakage integral; the floors of Section 1 in their Krylov form along the run; the validation split; and the Euler-versus-direct comparison at several Δ from a shared initialisation.

Hyperparameter transfer (negative result, n=200). A sweep over widths 48–384 and output scales 0.25–2 at fixed budget $3000/s^2$ did not bracket a training-error optimum (the richest scale was best at every width and depth) and the feature-to-lazy ratio of the descent rate did not collapse across widths under either the standard or the μP-like labelling; at fixed scale it grew with width. On training error at fixed budget, richness has no cost here, so the transfer question needs a criterion with an optimum (test error or a stability constraint) and is left open.

A The chain under truncation: error propagation and verifications

A.1 How a truncation at one stage reaches time T

A truncated chain replaces the exact stage map by its order-k expansion about the stage's initial state. Three exact structures decide how the local error of stage s appears at T.

Proposition 22 (The propagator is algebraic)**.** *A perturbation of the state at time s evolves by $\dot{\delta\theta} = -H(t)\,\delta\theta$, $H = J^\top J/n + \frac{1}{n}\sum_i E_i \nabla^2 N_i$, so the propagator $\Phi(T,s) = \mathrm{Texp}(-\int_s^T H)$ obeys $\|\Phi(T,s)\| \le \exp\big(\int_s^T \lambda_{\max}(-H)_+\big) \le \exp\big(\int_s^T \frac{1}{n}\|E\|H_2\big)$: only the residual-curvature part can expand, and its integrand decays with the residual. [n=200: $\|\Phi\| \le 21$ over a full run in the worst direction, i.e. an integrated negative curvature $\ln 21 \approx 3$, an average of*

10^{-3} *per unit time; the* H_2 *form of the bound is valid but loose by* e^{13}*, since it replaces the negative curvature by its worst case* $\|E\|H_2/n \approx 10^{-2}$*.]*

Proposition 23 (Three channels)**.** *A feature error* $\delta\Psi_s$ *at stage* s *reaches* T *as (i) an additive, undamped feature offset* $\Phi(T,s)\delta\Psi_s$ *(since* $\Psi_T = \Psi_0 + \sum_r \delta\Psi_r$*), hence a kernel error linear in the offset while it is smaller than the features; (ii) a residual error* $\delta E(T) = -\int_s^T \mathrm{Texp}(-\int_u^T K)\,\delta K(u)E(u)\,du$*, contracted by* K *in every lifted direction and sourced only by the kernel error acting on the residual; (iii) a feedback on later stages,* $\delta(\delta\Psi_r) \sim J'[J^\top \delta E]$ *for* $r > s$*, bounded by Proposition 22 and of the order of (ii) times the drive.*

Theorem 24 (Global error of the order-k chain)**.** *If each stage's order-*$(k{+}1)$ *remainder is at most* ε *times its first-order increment, then* $\|\Psi_T^{\mathrm{chain}} - \Psi_T\| \le \varepsilon\,\Phi_{\max}\sum_s \|\delta\Psi_s\|$ *and* $\|E_T^{\mathrm{chain}} - E_T\| \le \frac{2\varepsilon\Phi_{\max}}{n}\|\Psi_T\|\sum_s \|\delta\Psi_s\| \int_0^T \|E\| + O(\varepsilon^2)$*. The stage errors are proportional to the stage* increments *and add, so their sum is* ε *times the total feature change along the path, not* ε *times the number of stages: the chain is consistent of order* k *and stable with an algebraic propagator, hence convergent, uniformly in the number of stages. Fewer, longer stages at higher order and many short stages at first order both converge; they differ in cost, with* $S_k \sim \varepsilon^{-1/k}\int(\|\Psi^{(k+1)}\|/\|\Psi'\|)^{1/k}$ *stages needed.*

Proof. Triangle inequality over stages with Proposition 23(i), Proposition 22 for the propagator, variation of constants for (ii). The numerical convergence to three digits across a tenfold range of stage budgets at $k = 1$ is this theorem observed. □

Remark 25. For proofs the $k = 1$, many-stage end suffices: it is the flow itself, and nothing in this section needs more than first order per stage plus the propagator bound. For computation the chain is not cheap: long stages need high-order directional derivatives of the Jacobian at every expansion point, short stages need of order 10^4 of them; either way its cost is that of integrating the dynamics. Its value is the error accounting, which makes every truncated or reduced version of it a controlled approximation with a named error term. The constants are finite but not small in exactly two circumstances: early, where the residual-curvature part of H makes the propagator grow (the sharpness-reduction phase), and for errors landing in a slow mode, where (ii) gives no damping and the offset of (i) is later amplified by the lift of that mode. That is the chain's own statement of Propositions 19–20: what is wrong, or unknown, in the slow modes early survives and is amplified; what is wrong in the fast modes is forgotten.

A.2 Numerical verifications of the identities and of the leading orders

Remark 26. A numerical realisation of the chain (one eigendecomposition of the $N \times N$ kernel per stage, the maps of Proposition 7(c,d) applied to the network's own units) reproduces the real gradient-flow run to four digits in $\|E\|$ and to 0.1% in every kernel eigenvalue at $t = 8$, tracks it to 1–3% at T on every design tested including depth 8 and

width 512, and is converged to three digits across a tenfold range of stage budgets [n=200]. It is gradient flow on the sample integrated in large stages; its only use in this section is as the exact reference against which every reduced statement is checked.

Remark 27 (What was measured, n=200). (a) is verified to within 10% at initialisation for tanh, ReLU, GELU and softplus; the bulk is 2–5% of the core for tanh at unit gain and 30–50% for ReLU (whose local derivative is a 0/1 pattern), yet 98–100% of the first-order change lies in three singular directions for every activation. Along the path the independence of (a) fails within a few steps, as it must once the units adapt to the residual, while the rank persists: at $t = 0.3$ the per-unit-mean split gives a bulk ratio of 0.9 against the dispersion prediction of 0.05, and the cumulative change is still 98% in three directions; the correlation-free bound (b) holds with $r = 1$–2, i.e. the update lies 93% in 3 and 99.9% in 9 directions. Cores of successive stages add coherently while the drive keeps its direction, which is the cross-layer alignment measured earlier (randomising one layer's rotation destroys the lift); the bulk-plus-spike spectrum of trained weights is the random initialisation plus this coherent low-rank sum.

Remark 28 (Verification, n=200). With $\epsilon = 1/s$ and the coefficients through second order (including the second-order weight change and the feedback of the lifted kernel on the lazy residual), the predicted lift of init eigen-directions agrees with the measured one to 0–16% at output scales $s = 2, 4, 8, 16$ for as long as the measured lift is below order one, including the early *decrease* along deep-tail directions and its sign change; the truncated series fails sharply once the lift exceeds one, where Theorem 14 says it must. The leading coefficient a_1 changes sign along τ, so no universal power of s describes the delay to a given lift; the function is predicted, the exponent does not exist.

Remark 29 (What has not been measured at any sample size). The transported cutoff bracket of Section 2 with its leakage integral and the tracked decay rates; the per-stage sign condition as the smallest eigenvalue of the increment compressed to the residual's subspace; the floors of Section 1 in their Krylov form at a sample size with a genuine late stage; and the preconditioner term of the stage identity under Adam. Each is listed with its decision criterion in the experimental plan accompanying this draft.